

Binomial Theorem: Terms and Coefficients — Problem Set

Warm-up

1. Write down the expansion of $(1 + x)^4$ using Pascal's triangle.
2. State the general term T_{k+1} in the expansion of $(a + b)^n$.
3. Identify the number of terms in the expansion of $(3x - 2)^{12}$.
4. Write down the first three terms in the expansion of $(1 + 2x)^6$ in ascending powers of x .
5. Find the coefficient of x^2 in the expansion of $(1 - x)^5$.
6. Express nC_r in terms of factorials.
7. In the expansion of $(x + a)^n$, which term is the constant term if it exists?
8. Determine the power of x in the 4th term of $(x^2 + 3)^8$.
9. Find the coefficient of x^3 in $(2 + x)^4$.
10. Evaluate the term independent of x in $(x + \frac{1}{x})^2$.

Skill-building

1. Find the coefficient of x^3 in the expansion of $(2 + x)^7$.
2. Determine the term independent of x in the expansion of $(x + \frac{1}{x})^4$.
3. Find the middle term in the expansion of $(x - 3)^8$.
4. Calculate the ratio of the coefficient of x^2 to the coefficient of x^3 in $(1 + 4x)^{10}$.
5. Find the coefficient of x^5 in the expansion of $(2x - 3)^9$.
6. In the expansion of $(1 + kx)^5$, the coefficient of x^2 is 40. Find the value of k .
7. Find the term containing x^4 in the expansion of $(3x^2 - \frac{1}{x})^5$.
8. Prove that ${}^nC_r = {}^nC_{n-r}$ for any integers $n \geq r \geq 0$.
9. Find the coefficient of x^3 in the expansion of $(1 + x)^2(1 - x)^6$.
10. Find the term independent of x in the expansion of $(2x^3 - \frac{1}{x^3})^4$.

Easier Exam Questions

1. (Glenwood 2022 Q2)

What is the coefficient of x^4 in the expansion of $(2x - 1)^{10}$?

- (A) -210
- (B) 210
- (C) $1\,680$
- (D) $3\,360$

2. (Glenwood 2023 Q2)

Considering the expansion of $(1 + x)^7$, which statement is **not** true.

- (A) One of the terms is $21x^5$.
- (B) The constant term is 1.
- (C) The general term is of the form ${}^7C_k \cdot x^k$.
- (D) There are seven terms altogether.

3. (Glenwood 2019 Q9)

Find the coefficient of x^6 in the expansion of $(5x^2 - \frac{3}{x})^6$.

- (A) $-\binom{6}{4} \cdot 5^8 \cdot 3^2$
- (B) $-\binom{6}{4} \cdot 5^6 \cdot 3^6$
- (C) $\binom{6}{4} \cdot 5^8 \cdot 3^2$
- (D) $\binom{6}{4} \cdot 5^4 \cdot 3^2$

4. (North Sydney Boys 2023 Section E Q1)

Find the term in x^5 in the expansion of $(2x - \frac{1}{x^2})^8$.

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5. (Glenwood 2024 Q3)

What is the sixth term in the expansion of $(2x + 3)^8$?

- (A) ${}^8C_4(2x)^4(3)^4$
- (B) ${}^8C_4(2x)^3(3)^5$
- (C) ${}^8C_5(2x)^2(3)^7$
- (D) ${}^8C_5(2x)^3(3)^5$

6. (Blacktown Boys 2020 Q12d)

Find the constant term of $(4x^3 + \frac{3}{x})^8$.

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7. (Hurlstone 2019 Q8a)

Find the co-efficient of x in $\left(x + \frac{4}{x}\right)^5$ **2**

8. (Sydney Girls 2019 Q3a)

Find the term independent of x in the expansion $\left(x^2 + \frac{2}{x}\right)^{15}$. **2**

9. (North Sydney Boys 2022 Section C Q1)

Find the coefficient of x^2 in the expansion of $\left(x^2 + \frac{2}{x}\right)^7$ **2**

10. (SCEGGS 2023 Q10 a & b)

(a) Prove Pascal's identity, ${}^nC_k = {}^{n-1}C_{k-1} + {}^{n-1}C_k$ given that n and r are natural numbers, where $n > r$. **2**

(b) Find the term independent of x in the expansion $\left(2x^2 - \frac{1}{x}\right)^{12}$. **2**

11. (Glenwood 2020 Q7c)

Find the term independent of x in the binomial expansion of $\left(2x - \frac{1}{x^2}\right)^9$ **2**

12. (Gosford 2022 Q2d)

Find the coefficient of x^3 in the expansion of $\left(2x - \frac{1}{x}\right)^{11}$ **2**

Harder Exam Questions

1. (Baulkham Hills 2020 Q13c)

If $(1 + kx)^n = 1 - 12x + 60x^2 - \dots$, find the values of k and n . **3**

2. (Baulkham Hills 2023 Q7)

Which of the following is the coefficient of x^n in the expansion of $(1 + x)^n + (1 - x)^n$?

- (A) 0
- (B) $\frac{1+(-1)^n}{2}$
- (C) $1 + (-1)^n$
- (D) 2

3. (Knox 2021 Q13a & b)

Express $\frac{{}^{10}C_k}{{}^9C_k}$ in simplest form. **1**

(i) Expand $\left(x + \frac{2}{x}\right)^5$, giving your answer in simplest form. **1**

(ii) Hence simplify $\left(x + \frac{2}{x}\right)^5 + \left(x - \frac{2}{x}\right)^5$, again leaving your answer in simplest form. **1**

4. (Baulkham Hills 2019 Q11 a & b)

Express $\frac{{}^{10}C_k}{{}^9C_k}$ in simplest form **2**

Consider the expansion $\left(\frac{x^3}{2} + \frac{a}{x}\right)^8$. If the constant term is 5103, determine the possible value(s) of a . **3**

5. (Baulkham Hills 2020 Q10)

What is the coefficient of x^3 in the expansion of $(1 + x + x^2)^5$?

(A) 10

(B) 20

(C) 30

(D) 40

6. (Knox 2021 Q13d)

In the expansion of $(ax + b)^6$ where a and b are constants, the coefficient of x^4 is 2160 and the coefficient of x^5 is -576 . Find a and b . **3**

7. (North Sydney Boys 2022 Section C Q2)

Consider the expansion of $(1 + 2x)^n$

(i) Write down an expression for the coefficient of the term in x^4 **2**

(ii) The ratio of the coefficient of the term in x^4 to that of the term in x^6 is $5 : 8$. Find the value of n . **3**

8. (Girraween 2024 Q11b)

In the binomial expansion of $(1 + \frac{x}{k})^n$, the coefficient of x^3 is twice the coefficient of x^2 . Prove that $n = 6k + 2$ **2**

9. (Girraween 2023 Q112c)

In the expansion of $(5x + 2)^{20}$, the coefficients of x^k and x^{k+1} are equal.

Find the value of k . **3**